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9. If the linear momentum of the object is increased by 0.1%, then the kinetic energy is increased by

(a) 0.1 % (b) 0.2%

(c) 0.4% (d) 0.01%

10. If the potential energy of the particle is $\alpha - \frac{\beta}{2}x^2$, then force experienced by the particle is

(a) $F = \frac{\beta}{2}x^2$ (b) $F = \beta x$

(c) $F = -\beta x$ (d) $F = -\frac{\beta}{2}x^2$

11. A wind-powered generator converts wind energy into electric energy. Assume that the generator converts a fixed fraction of the wind energy intercepted by its blades into electrical energy. For wind speed v, the electrical power output will be proportional to

(a) v (b) v^2

(c) v^3 (d) v^4

12. Two equal masses m_1 and m_2 are moving along the same straight line with velocities $5\,ms^{-1}$ and $-9\,ms^{-1}$ respectively. If the collision is elastic, then calculate the velocities after the collision of m_1 and m_2, respectively

(a) $-4\,ms^{-1}$ and $10\ ms^{-1}$

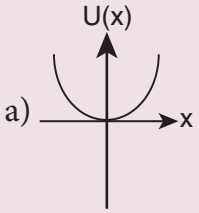
(b) $10ms^{-1}$ and $0\ ms^{-1}$

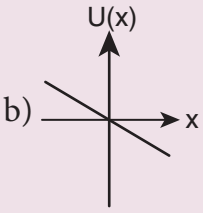
(c) $-9\,ms^{-1}$ and $5\ ms^{-1}$

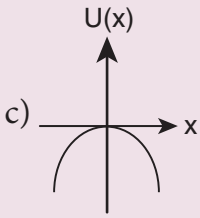
(d) $5\ ms^{-1}$ and $1\ ms^{-1}$

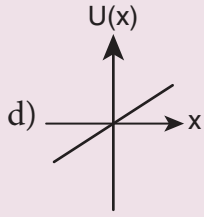
13. A particle is placed at the origin and a force $F = kx$ is acting on it (where k is a positive constant). If $U(0) = 0$, the graph of $U(x)$ versus x will be (where U is the potential energy function)

(IIT 2004)

a) 

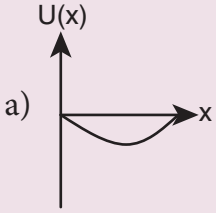
b) 

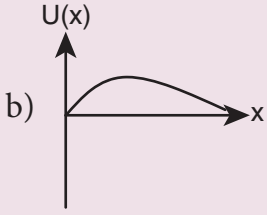
c) 

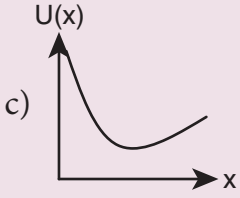
d) 

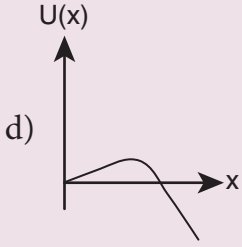
14. A particle which is constrained to move along x-axis, is subjected to a force in the same direction which varies with the distance x of the particle from the origin as $F(x) = -kx + ax^3$. Here, k and a are positive constants. For $x \geq 0$, the functional form of the potential energy $U(x)$ of the particle is

(IIT 2002)

a) 

b) 

c) 

d) 

15. A spring of force constant k is cut into two pieces such that one piece is double the length of the other. Then, the long piece will have a force constant of

(a) $\frac{2}{3}k$ (b) $\frac{3}{2}k$

(c) $3k$ (d) $6k$

Answers

1) c	**2)** d	**3)** a	**4)** a	**5)** b
6) a	**7)** c	**8)** b	**9)** b	**10)** c
11) c	**12)** c	**13)** c	**14)** d	**15)** b